

Brivalemer Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-853/A

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Cleaning verification swab results were below the health-based exposure limits derived for the product.

The control strategy links critical quality attributes to critical process parameters identified during development. No changes to the approved analytical procedures are proposed in this submission. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Cleaning verification swab results were below the health-based exposure limits derived for the product. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

3.2.P.4 Control of Excipients

For the commercial formulation F-853/A, FlowLac 100 (Lactose monohydrate, diluent) is used.

Lactose monohydrate complies with the requirements of ICH Q3D, ICH Q6A and USP-NF.

Selection Justification

The selection of FlowLac 100 is justified by reduced lot-to-lot variability reported by the qualified supplier observed during development studies.

Registered Specification

The current approved commercial specification (SPEC-EX-9104 Rev 05) lists Lactose monohydrate as FlowLac 100.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.